

Muhammad Awais

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EXPERIENCE

Software Engineering Intern — Full Stack

Jan 2026 – Mar 2026

National Incubation Center (NIC), Islamabad

- Engineered a production-grade tourism platform (stayOvers.pk) with AI-assisted booking workflows and strictly enforced RBAC for Guest, Host, and Admin personas.
- Architected a modular OOP backend, resolved critical bottlenecks via structured debugging, and maintained Git branching workflows with Postman API test suites.
- Participated in Agile sprint planning and daily stand-ups, collaborating with cross-functional teams to align technical deliverables with business goals.

Freelance Software Developer

May 2024 – Aug 2025

Fiverr — Remote, International Clients

- Designed and delivered responsive full-stack web applications for diverse international clients utilizing React, Next.js, and Python.
- Integrated RESTful APIs for secure payment processing and user authentication, optimizing backend queries to significantly reduce application load times.
- Managed end-to-end project lifecycles from architecture to deployment on Vercel and Netlify, meeting tight deadlines and maintaining high client satisfaction.

EDUCATION

National University of Modern Languages (NUML)

Jan 2022 – Jan 2026

Bachelor of Science in Computer Science

CGPA: 3.34/4.00

PROJECTS

Real-Time Traffic Choking Detection (FYP)

YOLOv8 · OpenCV · Python · Computer Vision

- Built a real-time congestion detection system using a **custom YOLOv8 model** trained on a **self-collected dataset** of local traffic videos and images; annotated data from scratch.
- Fine-tuned the detector to classify roads as free-flow or choked, and optimized the **OpenCV** inference pipeline for improved FPS via data augmentation and threshold tuning.
- Integrated the trained computer vision model into a web application, deploying the inference engine to enable real-time traffic monitoring and status alerts via an interactive dashboard.

AI-Powered Smart Health Platform [GitHub]

React.js · Node.js · Azure · REST APIs

- Built a health advisory application with a responsive **React.js** frontend (distributed as an **Android APK**) and a **Node.js** backend deployed on **Microsoft Azure**.
- Engineered secure **REST APIs** to process patient health data against clinical guidelines for real-time recommendations, ensuring high availability and seamless data flow.

ArticleSift — AI News Intelligence Platform [GitHub]

LLaMA 3.1 · Streamlit · OpenAI API · gTTS · Python

- Built an LLM-powered news Q&A application using **LLaMA 3.1** and the Google Custom Search API to aggregate and analyze real-time news data.
- Engineered an NLP backend with 5 endpoints (fetch, Q&A, summarize, audio, links), **gTTS** narration, and multi-language translation (English, Urdu, Spanish, French).

Multi-Agent Research Automation System [GitHub]

LangChain · LLMs · MCP · Python · Generative AI

- Engineered a **multi-agent research pipeline** using **LangChain** and LLMs to autonomously generate end-to-end research reports.
- Implemented an **Outline Agent** for structural planning and a **Writer Agent** for content expansion, coordinated via the **Model Context Protocol (MCP)** for orchestration.

SKILLS

- Languages & Databases:** JavaScript, TypeScript, Python, SQL.
- Frameworks & Tools:** React.js, Node.js, Next.js, FastAPI, REST APIs, LangChain, Git, GitHub, Postman, Docker.
- AI & ML:** Machine Learning, NLP, Generative AI, LLMs, Agentic AI, Computer Vision, YOLOv8.
- Cloud & Deployment:** Microsoft Azure, Vercel, Netlify, CI/CD.

CERTIFICATIONS

Agentic AI Level 2 Developer — PIAIC

MCP Automation in Production — Hugging Face

PCAP: Programming Essentials in Python — Cisco

Generative AI: Prompt Engineering — IBM

Essentials of Generative AI — Pak Angels

SQL Certification — DataCamp

IT Essentials — Cisco

Machine Learning Specialization — DeepLearning.AI

Computer Vision & YOLO Object Detection

Postman API Fundamentals Expert

Introduction to Git

Google Soft Skills Programme — Google